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Subject Name: Database Management System

Week 5

Module 4 : SQL

Faculty Name : UTTAM WAGHMODE

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Lecture 13 –HAVING ,GROUPBY, NULL

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Lecture No: 13: HAVING ,GROUPBY, NULL



HAVING- it places condition on groups created by GROUP BY

Emp_ID	Name	Age	Address	Salary	Dept_ID
1	Rahul	32	Delhi	2000.00	500
2	Kamal	25	Pune	1500.00	500
3	Karan	23	Pune	2000.00	500
4	Chirag	25	Mumbai	6500.00	501
5	Harsh	32	Mumbai	8500.00	501
6	Kajal	22	Bilaspur	4500.00	502
7	Mahi	24	Patna	10000.00	502

Write a query to display departments where the number of employee is greater than 2

```
SELECT Dept_id, COUNT(*)
FROM Emp
GROUP BY Dept_id
HAVING COUNT(*) >2;
```



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- **OUTPUT:**

Dept_ID	COUNT(*)
500	3

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Emp_ID	Name	Age	Address	Salary	Dept_ID
1	Rahul	32	Delhi	2000.00	500
2	Kamal	25	Pune	1500.00	500
3	Karan	23	Pune	2000.00	500
4	Chirag	25	Mumbai	6500.00	501
5	Harsh	32	Mumbai	8500.00	501
6	Kajal	22	Bilaspur	4500.00	502
7	Mahi	24	Patna	10000.00	502

Write a query to find the department for which the sum of salaries of employees is more than 10000

```
SELECT Dept_ID, SUM(Salary)
FROM Emp
GROUP BY Dept_id
HAVING SUM(Salary)>10000;
```



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- **OUTPUT:**

Dept_ID	SUM(Salary)
501	15000.00
502	14500.00

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Emp_ID	Name	Age	Address	Salary	Dept_ID
1	Rahul	32	Delhi	2000.00	500
2	Kamal	25	Pune	1500.00	500
3	Karan	23	Pune	2000.00	500
4	Chirag	25	Mumbai	6500.00	501
5	Harsh	32	Mumbai	8500.00	501
6	Kajal	22	Bilaspur	4500.00	502
7	Mahi	24	Patna	10000.00	502

Write a query to find the sum of salaries of employee that lives at same location

```
SELECT Address, SUM(Salary)
FROM Emp
GROUP BY Address
HAVING COUNT(Address)>=2;
```



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- **OUTPUT:**

Address	SUM(Salary)
Mumbai	15000.00
Pune	3500.00

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NULL VALUES

- ***null*** signifies
  - **an unknown value/missing**, or
  - **a value that does not exist**

Emp_Id	Name	Phone_No	Passport_No
001	Rahul	7777777777	111 111 111
002	Anil	8888888888	NULL



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- The result of any **arithmetic expression** (+, -, *, /) involving ***null*** must return a ***null***.
  - *Eg: 5 + null = null*

    *null * 5 = null*
- **Aggregate functions** simply **ignore** **null values** (as in SQL)
  - Eg: Aggregate functions ***avg, min, max, sum*** ignores *null values* **except** for ***count***
- For **duplication, elimination** and **grouping**, **null** is ***treated like any other value***, and two nulls are assumed to be the same (as in SQL)

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- Result of any **arithmetic expression** involving ***null*** is ***null***
- Result of **where** clause condition is ***false*** if it evaluates to ***null***.
- **and, or, not operators** handles ***null*** as follows:

and	true	false	null
true	true	false	null
null	null	false	null
false	false	false	false

or	true	false	null
true	true	true	true
null	true	null	null
false	true	false	null

not	
true	false
null	null
false	true

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Go

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HOW TO TEST NULL VALUES ?

- TESTING NULL VALUES WITH COMPARISON OPERATORS, SUCH AS = OR <> , ARE NOT POSIBLE.
- WE CAN USE ***IS NULL*** AND ***IS NOT NULL*** OPERATORS.

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Table Name = emp

empno	ename	job	mgr	hiredate	sal	comm	deptno
7369	SMITH	CLERK	7902	1993-06-13	800.00	0.00	20
7499	ALLEN	SALESMAN	7698	1998-08-15	1600.00	300.00	30
7521	WARD	SALESMAN	7698	1996-03-26	1250.00	500.00	30
7566	JONES	MANAGER	7839	1995-10-31	2975.00		20
7698	BLAKE	MANAGER	7839	1992-06-11	2850.00		30
7782	CLARK	MANAGER	7839	1993-05-14	2450.00		10
7788	SCOTT	ANALYST	7566	1996-03-05	3000.00		20
7839	KING	PRESIDENT		1990-06-09	5000.00	0.00	10
7844	TURNER	SALESMAN	7698	1995-06-04	1500.00	0.00	30
7876	ADAMS	CLERK	7788	1999-06-04	1100.00		20
7900	JAMES	CLERK	7698	2000-06-23	950.00		30
7934	MILLER	CLERK	7782	2000-01-21	1300.00		10
7902	FORD	ANALYST	7566	1997-12-05	3000.00		20
7654	MARTIN	SALESMAN	7698	1998-12-05	1250.00	1400.00	30

DISPLAY THE NAME OF EMPLOYEES HAVING COMMISSION (COMM) AS NULL VALUE.

```
SELECT ename
FROM emp
WHERE comm IS NULL
```



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Emp Table

Emp_ID	Name	Age	Address	Salary	Email
1	Rahul	32	Delhi	2000.00	rr@gmail.com
2	Kamal	25	Pune	1500.00	kk@gmail.com
3	Karan	23	Pune	2000.00	NULL
4	Chirag	25	Mumbai	6500.00	cc@gmail.com
5	Harsh	32	Mumbai	8500.00	hh@gmail.com
6	Kajal	22	Bilaspur	4500.00	NULL
7	Mahi	24	Patna	10000.00	mm@gmail.com

List employees who have no associated email Ids

```
SELECT *
FROM Emp
WHERE Email IS NULL;
```



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- **OUTPUT:**

Emp_ID	Name	Age	Address	Salary	Email
3	Karan	23	Pune	2000.00	NULL
6	Kajal	22	Bilaspur	4500.00	NULL

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Emp Table

Emp_ID	Name	Age	Address	Salary	Email
1	Rahul	32	Delhi	2000.00	rr@gmail.com
2	Kamal	25	Pune	1500.00	kk@gmail.com
3	Karan	23	Pune	2000.00	NULL
4	Chirag	25	Mumbai	6500.00	cc@gmail.com
5	Harsh	32	Mumbai	8500.00	hh@gmail.com
6	Kajal	22	Bilaspur	4500.00	NULL
7	Mahi	24	Patna	10000.00	mm@gmail.com

List employees who have associated email Ids

```
SELECT *
FROM Emp
WHERE Email IS NOT NULL;
```



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- **OUTPUT:**

Emp_ID	Name	Age	Address	Salary	Email
1	Rahul	32	Delhi	2000.00	rr@gmail.com
2	Kamal	25	Pune	1500.00	kk@gmail.com
4	Chirag	25	Mumbai	6500.00	cc@gmail.com
5	Harsh	32	Mumbai	8500.00	hh@gmail.com
7	Mahi	24	Patna	10000.00	mm@gmail.com

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Lecture No: 14: IN, SUBQUERY, BETWEEN



IN OPERATOR

- The **IN** operator allows us **to specify multiple values in a WHERE clause.**
  - It allows us to easily test if an expression matches any value in a list of values
- The **IN** operator is **a shorthand for multiple OR conditions.** So it **used to replace many OR conditions.**

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Emp_ID	Name	Age	Address	Salary
1	Rahul	32	Delhi	2000.00
2	Kamal	25	Pune	1500.00
3	Karan	23	Pune	2000.00
4	Chirag	25	Mumbai	6500.00
5	Harsh	32	Mumbai	8500.00
6	Kajal	22	Bilaspur	4500.00
7	Mahi	24	Patna	10000.00

Example 1:

It selects all Employees located in Delhi, Bilaspur and Patna

```
SELECT *
FROM Emp
WHERE Address IN ('Delhi', 'Bilaspur','Patna);
```



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- **OUTPUT :**

Emp_ID	Name	Age	Address	Salary
1	Rahul	32	Delhi	2000.00
6	Kajal	22	Bilaspur	4500.00
7	Mahi	24	Patna	10000.00

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SUBQUERY USING IN OPERATOR

Emp_ID	Name	Age	Address	Salary
1	Rahul	32	Delhi	2000.00
2	Kamal	25	Pune	1500.00
3	Karan	23	Pune	2000.00
4	Chirag	25	Mumbai	6500.00
5	Harsh	32	Mumbai	8500.00
6	Kajal	22	Bilaspur	4500.00
7	Mahi	24	Patna	10000.00

Example 3:

It selects all employees that are from the same address as the customers

```
SELECT *
FROM Emp
WHERE Address IN(SELECT Address FROM Customers);
```



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CUSTOMERS ADDRESS: SUBQUERY

```
SELECT *
FROM Emp
WHERE Address IN(SELECT Address FROM Customers);
```

Address
Delhi
Pune
Pune
Mumbai
Mumbai
Bilaspur
Patna

Address
Delhi
Bhopal
Bangaluru
Chennai
Bilaspur

EMPLOYEES ADDRESS: MAIN QUERY

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- **OUTPUT :**

Emp_ID	Name	Age	Address	Salary
1	Rahul	32	Delhi	2000.00
6	Kajal	22	Bilaspur	4500.00

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BETWEEN

Emp_ID	Name	Age	Address	Salary
1	Rahul	32	Delhi	2000.00
2	Kamal	25	Pune	1500.00
3	Karan	23	Pune	2000.00
4	Chirag	25	Mumbai	6500.00
5	Harsh	32	Mumbai	8500.00
6	Kajal	22	Bilaspur	4500.00
7	Mahi	24	Patna	10000.00

It selects all Employees with the Salary between 2000 to 7000

```
SELECT *
FROM Emp
WHERE Salary BETWEEN 2000 AND 7000;
```



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- **OUTPUT :**

Emp_ID	Name	Age	Address	Salary
1	Rahul	32	Delhi	2000.00
3	Karan	23	Pune	2000.00
4	Chirag	25	Mumbai	6500.00
6	Kajal	22	Bilaspur	4500.00

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Lecture No: 15: LIKE, WILDCARD..%,_



LIKE OPERATOR WITH WILDCARD

Emp_ID	Name	Age	Address	Salary
1	Rahul	32	Delhi	2000.00
2	Kamal	25	Pune	1500.00
3	Karan	23	Pune	2000.00
4	Chirag	25	Mumbai	6500.00
5	Harsh	32	Mumbai	8500.00
6	Kajal	22	Bilaspur	4500.00
7	Mahi	24	Patna	10000.00

It selects all Employees with a *Name* that have "a" in second position

```
SELECT * FROM Emp
WHERE Name LIKE "_a%";
```


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- **OUTPUT :**

Emp_ID	Name	Age	Address	Salary
2	Kamal	25	Pune	1500.00
3	Karan	23	Pune	2000.00
5	Harsh	32	Mumbai	8500.00
6	Kajal	22	Bilaspur	4500.00
7	Mahi	24	Patna	10000.00

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Emp_ID	Name	Age	Address	Salary
1	Rahul	32	Delhi	2000.00
2	Kamal	25	Pune	1500.00
3	Karan	23	Pune	2000.00
4	Chirag	25	Mumbai	6500.00
5	Harsh	32	Mumbai	8500.00
6	Kajal	22	Bilaspur	4500.00
7	Mahi	24	Patna	10000.00

It selects all Employees with a *Name* starting with "M" and are at least 4 characters in length

```
SELECT * FROM Emp
WHERE Name LIKE "M_ _ _%";
```



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- **OUTPUT :**

Emp_ID	Name	Age	Address	Salary
7	Mahi	24	Patna	10000.00

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Examples : LIKE operator with wildcard operator (% , _):

LIKE Operator	Description
WHERE Name LIKE 'A%'	Finds any values that start with "**A**", Eg: Aman, Aakriti
WHERE Name LIKE '%a'	Finds any values that end with "**a**", Eg: Neha, Soma
WHERE Name LIKE '%oh%'	Finds any values that have "**oh**" in any position, Eg: Sohan, Mohan
WHERE Name LIKE '_r%'	Finds any values that have "**r**" in the second position, Eg: Priya, Priyesh
WHERE Name LIKE 'K_ _ _ _%'	Finds any values that start with "**K**" and are at least **5** characters in length, Eg: Kamal, Karan, Kajal
WHERE Name LIKE 'A_ _%'	Find, Eg: s any values that start with "**A**" and are at least **3** characters in length, Eg: Aki, Anu
WHERE Name LIKE 'A%N'	Finds any values that start with "**A**" and ends with "**N**", Eg: Aman, Raman

Activate Windows

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Thank You



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